

EZHILAN CHINNASAMY

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LinkedIn | GitHub | Portfolio

Data engineer combining hands-on platform delivery with cross-team leadership in payments. Designs data lakes, integrations and automation that make financial operations more reliable and analytics easier to use. Brings experience processing 2M records daily and translating business requirements into delivery across three teams.

SKILLS

- **Data Engineering:** ETL/ELT design, star-schema modelling, Airflow orchestration, data lakes, API/SFTP ingestion
- **Languages & Platforms:** SQL, Python, PostgreSQL, FastAPI, Azure Data Factory, AWS S3/EMR, PySpark
- **Analytics & Quality:** Financial reconciliation, evaluation harnesses, Grafana business-operations monitoring
- **Engineering Delivery:** Agile/Scrum, Git/GitHub, automated-test workflows, AI coding assistants

PROFESSIONAL EXPERIENCE

Pay with Spire, Business Intelligence Analyst Sep 2024 - Jun 2026 | Dallas, Texas, US

- Designed a bronze/silver/gold Azure data lake with Azure Data Factory pipelines processing 2M records daily for multi-team reporting, marketing automation and AI analytics.
- Led three-team delivery of settlement automation, translating payment-processing rules into domain architecture and coordinating vendor API/SFTP integration across operations and engineering.
- Built modular Python merchant onboarding with partner-file validation, API integration and regression tests, saving an estimated 8.5 hours of manual entry per week.
- Delivered three scheduled ADF pipelines for Zendesk, including incremental loads, moving API-based reporting to SQL and processing 500–1,200 changed tickets per daily run.
- Automated partner billing with Python and SQL, checking transaction counts, amounts and fees before generating and distributing monthly reports.
- Built a production multi-agent analytics layer over the warehouse using Google ADK and Vertex AI, enabling self-service NL2SQL/BQML querying and offloading approximately 30% of data-team workload.

The University of Texas at Dallas, Student Assistant Jun 2024 - Sep 2024

- Automated public-record integration and trained 50+ student researchers, accelerating project delivery by 30%.
- Consolidated 100k+ fragmented data points spanning five decades of Texas tech history into a timeline application adopted across departments for research.

Ernst & Young Global Delivery Services, Software Engineer Feb 2021 - Oct 2022 | Chennai, India

- Increased batch throughput by 24% with PySpark pipelines on AWS EMR processing multi-terabyte logs.
- Architected a secure migration framework from legacy multi-tenant databases to an AWS S3 data lake, strengthening IAM controls and improving query performance by 35%.

PROJECTS

Multi-Agent Financial Reconciliation Engine

- Implemented a reconciliation prototype matching ledger and settlement records without a shared transaction identifier, combining deterministic matching, LangGraph agents and SQL-backed MCP investigation tools.
- Shipped a ground-truth evaluation harness that caught two regressions; after finding inverted model confidence, added a deterministic dollar threshold to human-review routing.

Fintech Hybrid RAG Engine

- Engineered an ACH documentation Q&A service using BM25 and PostgreSQL/pgvector retrieval to produce source-cited answers to payments operations questions.
- Diagnosed retrieval failures with RAGAS evaluation, improving context recall from 0.410 to 0.603 (47% relative gain) through corpus filtering, query-aware ranking and explicit return-code inclusion.

EDUCATION & CERTIFICATIONS

The University of Texas at Dallas, MS in Business Analytics and Artificial Intelligence Jan 2023 - Dec 2024 | GPA: 3.7

Certifications issued by UTD: Applied Machine Learning Certificate

Kumaraguru College of Technology, Bachelors in Computer Science and Engineering Jul 2017 - Jun 2021 | GPA: 8.1

Scrum Master Accredited Certification - International Scrum Institute | Mar 2026